
FNH 200 942

Lesson 12 - Foodborne Illnesses

2026 Summer

Please recall a recent foodborne illness someone you know experienced:

What caused it?

What are some symptoms, please keep this professional :) ?

Team Project

Please add to the wiki, kindly indicate if your project is ready to be grade on the right column.

Team Number ↕	Product Investigated and Link ↕	Completed? To be updated by Aug 13 ↕
0 (Example)	The Processing and Safety of Swiss Cheese (Please provide link to the page generated above)	No
1	Course:FNH200/Projects/2026/Nescafe Coffee Pods/Capsules - What is inside?	No
2	Bear root (endangered food): Course:FNH200/Projects/2026/Bear root (endangered food)	No
3	Ice Cider (endangered food): Course:FNH200/Projects/2026/Ice Cider - Exploring the Endangered Drink	No
4	Course:FNH200/Projects/2023/Olestra Regulation	No
5	- If you are enhancing an existing page, add that existing link here - If you are doing an interview or researching an endangered food, create a new link using the Create Your Project Page tool below and add the link here	No
6	Pu-erh Tea (Endangered Food) Course:FNH200/Projects/2026/Pu-erh Tea	No
7	- If you are enhancing an existing page, add that existing link here - If you are doing an interview or researching an endangered food, create a new link using the Create Your Project Page tool below and add the link here	No



Course Overview:

There are 13 lessons in FNH 200, in varying lengths.

Foundation

Lesson 01

Food Science and Canadian Food System



Lesson 02

Chemical and Physical Properties of Foods



Lesson 03

Fat and Sugar Substitutes



Lesson 04

Food Standards, Regulations, and Guides



Preservation

Lesson 05

Lesson 06

Thermal



Lesson 07

Low Temperature



Lesson 08

Dehydration



Lesson 09

Biotechnology



Lesson 10

Irradiation



Healthy & Safe Future

Lesson 11

Affects on Nutrient Retention



Lesson 12

Toxicants in Foods and Foodborne Diseases



Lesson 13

Trends in Food for Nutrition and Health



Learning Objectives

- define **toxicity, hazard and risk**
- explain the importance of the **dose-response relationship**
- ~~outline the process for risk assessment~~
- describe examples of **naturally occurring (constituent or contaminant) and environmental toxicants** in the food supply
- assess **your** risk of food intoxication
- explain the relative importance of various factors as contributors to the **incidence of foodborne disease** in Canada;
- gain some insight into the major foodborne disease
- assess the potential risk of contracting food borne disease **at home**



Toxicity, Hazard, and Risk

Toxicity: had 'bad/toxic' a substance is

Hazard: probably of contacting some toxic

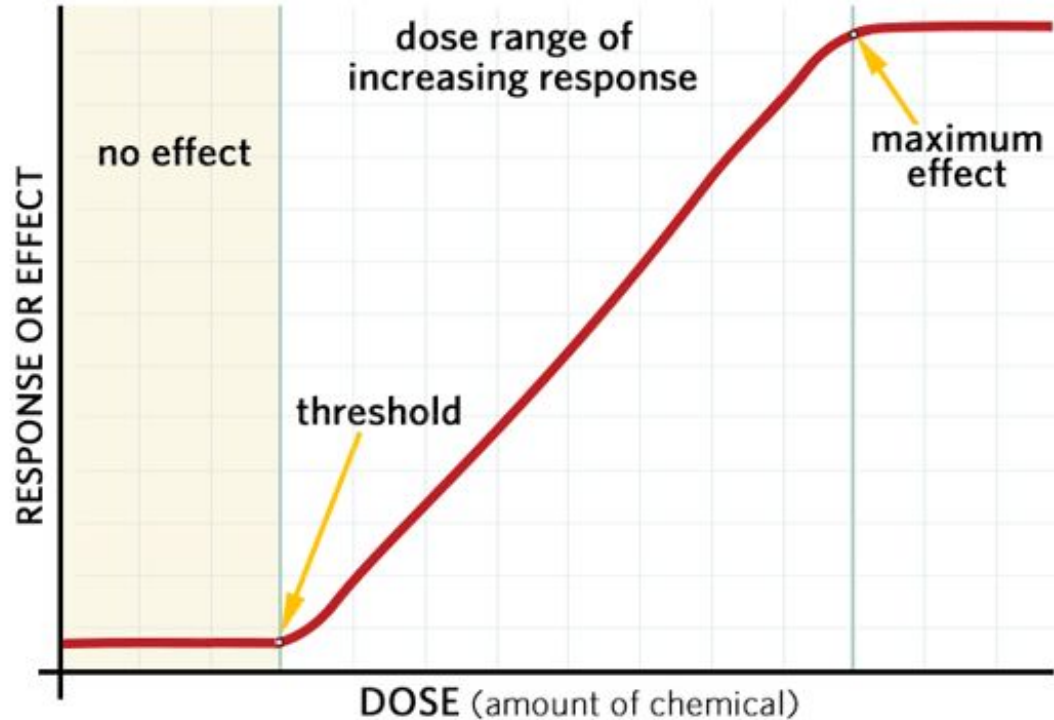
Risk: how likely a hazard will happen and the impact

Perceived Importance of Food Components

Food Component	Relative Hazard Ranking	
	Popular Notion	Regulatory Concern
Food Additives	Great	Minimal
Agricultural Chemicals		
Natural Toxicants		
Environmental Contaminants		
Nutritional Properties		
Microbiological Contaminants	Minimal	Great

Lesson 4: Dose Response Curve

What is a poison?



Naturally Occurring Toxicants: <i>Constituents</i>	Naturally Occurring Toxicants: <i>Contaminants</i>	Environmental Toxicants
Cholinesterase inhibitors: • solanine in potatoes Cyanogenic glycosides: • amygdalin in apple seeds Glucosinates: • Sinigrin in cabbage Protease inhibitors: • protease inhibitors in soy beans Nitrites: • Green leafy plants Allergens: • beta-lactoglobulin in milk, peanut proteins, soy proteins	Mycotoxins: • Aflatoxin • Patulin • Ochratoxin A • Vomitoxin • Zearalenone Examples: aflatoxin in mouldy peanuts, vomitoxin in wheat Bacterial toxins: • <i>botulinum</i> toxin • Staphylococcal toxin Seafood Toxins: • Histamine • Saxitoxin • Domoic acid • Tetrodotoxin	Food packaging residues: • Monomers of plastic resins Pesticide residues: • Herbicides • Insecticides • Fungicides Example: malathion residues on fruit Heavy metals: • Lead • Mercury • Cadmium Examples: lead in vegetables; mercury in large ocean fish (eg. swordfish) Animal drugs: • Antibiotics • Hormones Examples: penicillin in milk Radioisotopes: • from soil or • from radioactive fallout

Allergens, such as nuts, are:

- A. Natural Constituents
- B. Natural Contaminants
- C. Environmental Toxicants



Toxins found on mouldy nuts are known as:

- A. Natural Constituents
- B. Natural Contaminants
- C. Environmental Toxicants



Solanine - Green Potatoes

Normal Level - 2 to 13 mg per 100 gram of potato

Green part - 80 to 100 mg

Heat resistant

Drowsiness, vomiting and diarrhea, sensitivity,
difficulty in breathing

Death

Minimize risk: farming and storage practices



Name the toxin natural found in almonds.



- A. Cholinesterase inhibitors
- B. Amygdalin, cyanogenic glycoside
- C. Protease inhibitors
- D. Nitrates
- E. Allergens



Health Canada reminds Canadians about the risks of consuming apricot kernels

Starting date:	July 25, 2019
Posting date:	July 25, 2019
Type of communication:	Information Update
Subcategory:	Cyanide Poisoning
Source of recall:	Health Canada
Issue:	Food Safety
Audience:	General Public
Identification number:	RA-70545

Report a Concern

Last updated: 2019-07-25
July 25, 2019
For immediate release

<https://www.healthycanadians.gc.ca/recall-alert-rappel-avis/hc-sc/2019/70545a-eng.php>

“We can trim off the tiny mould grown on the food.” It is true, and depends on...

(i) Storage temperature

(ii) Texture of the food

(iii) Size of the mould grown

A. (i)

B. (ii)

C. (i) and (ii)

D. (ii) and (iii)

E. (i), (ii), and (iii)

The meltdown of the Fukushima Daiichi Power Facility after the tsunami in Japan in March 2011 introduces this type of toxicant in the food chain.

- A. Bacterial
- B. Constituent
- C. Contaminant
- D. Environmental
- E. Packaging residues



Examples of Environmental Toxicants

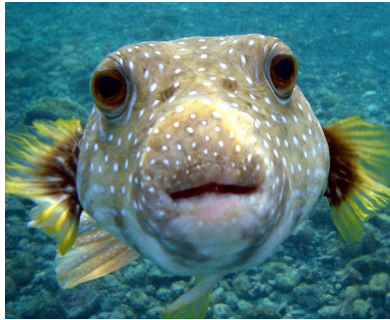
- Industrial activity
 - mercury, lead, dioxins, polychlorinated biphenyls, radioisotopes
 - Agricultural activity
 - pesticide and herbicide residues, antibiotic residues
 - Food processing
 - packaging residues, cleaning supplies, equipment maintenance
 - Naturally occurring environmental toxicants
 - mercury, radioisotopes
-

Name the neurotoxin that was first discovered on Prince Edward Island. Severely affected individual suffered from memory loss.

- A. Histamine
- B. Saxitoxin
- C. Domoic acid
- D. Tetrodotoxin



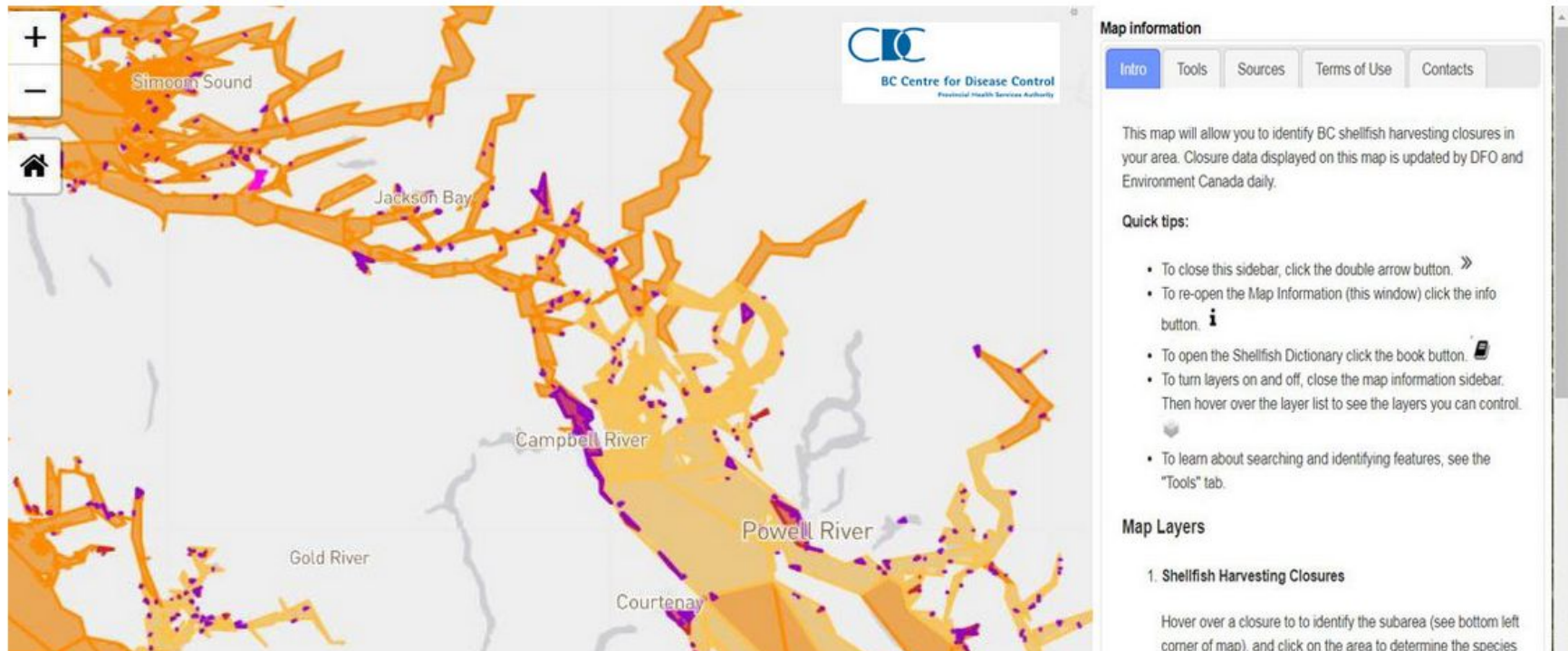
Cory Correia - CBC News - Posted: Jul 18, 2018 2:58 AM PT | Last Updated: July 18, 2018



<https://www.pac.dfo-mpo.gc.ca/fm-gp/contamination/index-eng.html>

Noctiluca scintillans has been spotted along coastal B.C. over the past month, including in the community of Saltair. (Murray Welte)

Shellfish Harvesting Status Map



Food Recall

Various Salem Foods brand Ground Spices and Spice Blends recalled due to undeclared wheat, sesame and mustard

Recall Food recall warning | 2026-06-26

Momchipz brand Veggie Chips (Broccoli Florets & Cauliflower) recalled due to undeclared gluten

Recall Food recall warning | 2026-08-07

Kehar brand Californian Pistachio Kernel recalled due to Salmonella

Recall Food recall warning | 2026-08-06

Heeva Fine Foods brand Pistachio Nuts recalled due to Salmonella

Recall Food recall warning | 2026-04-28

▼ Audience

General public 10343

Healthcare 4095

Industry 460

▼ Category

Vehicles 8073

Health product 4392

Consumer product 1346

Food 1087

▼ Issue

(-) Food

Microbial Contamination 189

Allergen 148

Chemical 4

Extraneous Material 4

Marine Biotoxin 1

▼ Recall class

Class 2 372

Class 1 346

Class 3 175

-- 2

Foodborne Illnesses

- Microorganisms
 - Parasites
 - Plants
 - Animals
 - Chemicals
-

A situation in which two or more people experience a similar illness after ingestion of the same food and where epidemiological evidence implicates the food as the source of the illness

- A. An incident
- B. A single case
- C. An outbreak

Maple Leaf Foods 2008
Outbreak
57 confirmed cases
22 deaths

Cold cut, listeriosis

Calgary · Updated

Calgary company at centre of daycare E. coli outbreak pleads guilty to bylaw offences

Charges to be dropped against Fueling Minds directors Faisal Alimohd and Anil Karim



[Meghan Grant](#) · CBC News · Posted: Apr 24, 2025 4:00 AM PDT | Last Updated: April 24



Food Intoxication

Organism	Symptoms onset	Common foods	Prevention
<i>Clostridium botulinum</i>	Botulinal toxins are among the most toxic substances known. Estimated LD₅₀ for humans is 1 ng/kg body weight. Dizziness, double vision, difficult swallowing, respiratory or cardiac paralysis, death (12-36 hrs)	Low acid foods in anaerobic conditions. Improperly home-canned, vacuum-packed, poorly refrigerated foods Baked potatoes wrapped in foil Spores have been detected in vegetables such as carrots, potatoes and fruit (contamination with soil), in fresh and processed meats and in honey and corn syrup.	Commercially sterile after processing must receive a "botulinum cook" (12 D) Use sodium nitrite for cured meats Properly refrigerate vegetable/fruit juice. Honey, corn syrup and molasses must not be fed to infants < 1 year of age. Pasteurization of honey does not inactivate the spores.

Organism	Symptoms onset	Common foods	Prevention
<i>Escherichia coli</i> O157:H7 (Shiga toxin)	Bloody diarrhea, kidney failure and hemolytic uremic syndrome (HUS) in sever cases (12-72 hrs)	Undercooked ground meat, raw milk, unpasteurized apple juice/cider, lettuce, spinach, alfalfa sprouts Contaminated water supply (outbreak in Walkerton, Ontario)	Practice good food sanitation and hand washing. Properly handle and cook foods.<i>E. coli</i> is killed by normal cooking and pasteurization, as well as ionizing radiation, but is able to survive in acidic conditions (e.g. apple juice). It does not grow well at temperatures below 8°C. The Kidney Foundation of Canada B.C. branch: E. coli bacteria: What You need to Know ↓

Food Infections

Organism	Symptoms onset	Common foods	Prevention
<i>Salmonella</i> spp.	Nausea, fever, vomiting, abdominal cramps, diarrhea (6-48 hrs)	Raw meats and poultry, eggs (about 60% of chickens sold in the retail market are contaminated with <i>Salmonella</i>), milk, dairy products	Salmonella are easily killed by heat (normal cooking and pasteurization processes) and ionizing energy, but survive freezing and dehydration. Consequently, frozen and thawed poultry can be just as contaminated as fresh poultry.

Organism	Symptoms onset	Common foods	Prevention
<i>Listeria monocytogenes</i>	Flu-like symptoms, meningitis, septicemia, or pneumonia. Birth defects, still birth (1 day- 3 weeks)	Raw milk, dairy products, vegetables, fish and meat products	The bacteria is capable of growing slowly on foods during storage in the refrigerator, and appears to be fairly tolerant to salt. It is easily killed by proper cooking and pasteurization techniques. Refrigerate raw materials and high-risk foods below 4°C.

Virus

Organism	Symptoms onset	Common foods	Prevention
Hepatitis A	Fever, nausea, vomiting, abdominal pain, fatigue, swelling of the liver, jaundice (10-50 days)	Foods prepared with human contact; contaminated water	Wash hands and practice good personal hygiene, avoid raw seafood

Organism	Symptoms onset	Common foods	Prevention
Noravirus	Vomiting, diarrhea, headache, fever (24-48 hrs)	Sewage, contaminated water, contaminated salad ingredients, raw seafood, infected food workers	Use potable water, practice good personal hygiene, cook all seafood

How to protect yourself?

Precautionary steps you may take to avoid foodborne illnesses.

What can be done to ensure food safety?

The World Health Organization (WHO) and the BC Health Services have developed ten rules for safe food handling, that if followed, can reduce the risk of foodborne disease.

Below are some key points that need to be considered

- Choose foods processed for safety (i.e. pasteurized products over raw)
- Cook foods thoroughly
- Eat cooked foods immediately
- Store food promptly and carefully. Perishable and "cold" foods should be kept at 4°C or below and reheated to 60°C or above.
 - Avoid the **"Temperature Danger Zone" (TDZ)** which is from 4°C to 60°C (40-140°F). Bacteria can grow and/or produce toxins in food if left in the TDZ.
- Reheat cooked foods thoroughly
- Avoid cross-contamination. Do not allow any contact between raw food or its traces and cooked food
- Wash hands repeatedly
- Keep all kitchen surfaces clean
- Do not let anyone with diarrhea or infected sores prepare food
- Wash all fruits and vegetables before eating
- Finally, **if in doubt, throw it out!** (Ministry of Health Services, 2001)

In the summer section, when most of you aren't going to become a food scientist, technologist, inspectors, etc, I care if you can apply these in your daily life more than concepts introduced on the following few slides.

What about the food industry?

HACCP - Hazard Analysis Critical Control Point

Created by NASA to ensure our astronauts' foods are safe

1. Identify Hazards
 2. Determine the critical control points (CCPs)
 3. Establish control measures (critical limits or thresholds)
 4. Establish procedures to monitor CCPs
 5. Establish corrective actions
 6. Keep records
 7. Verify procedures
-

HACCP of a canned food

Hazard Analysis Critical Control Points

1. Identify Hazards - *C. botulinum* spores
2. Determine the critical control points (CCPs) - canning
3. Establish control measures (critical limits or thresholds)
- temp, time
4. Establish procedures to monitor CCPs
- thermometer at cold spot, data logging
5. Establish corrective actions
- calculate additional time needed
6. Keep records
7. Verify procedures



Which of the following are examples of CCP?

- (i) Allergenic
 - (ii) Biological
 - (iii) Chemical
 - (iv) Environmental
 - (v) Physical
-

Keeping yourself, family, and community safe...

